Practice Questions on the Ricardian Model

Problem 1: Comparative Advantage and Opportunity Cost

Two countries, **A** and **B**, produce two goods: **Wheat** and **Cloth**. The labor productivity (output per worker per hour) is given below:

Country	Wheat (units per hour)	Cloth (units per hour)
A	10	5
B	4	2

(a) Calculate the opportunity cost of producing one unit of **wheat** in both countries.

(b) Calculate the opportunity cost of producing one unit of **cloth** in both countries.

(c) Which country has a **comparative advantage** in wheat production? Which has a comparative advantage in cloth production?

Problem 2: Gains from Specialization and Trade

Consider two countries, **X** and **Y**, producing **Cars** and **Computers**. The total labor force is **1,000 workers** in each country. The labor productivity is:

Country	Cars per worker per year	Computers per worker per year
X	4	8
Y	2	6

(a) Calculate the opportunity cost of producing **one car** in terms of computers in each country.

(b) If each country fully specializes in the good in which it has a comparative advantage, how many total **cars** and **computers** will be produced?

(c) Show that trade allows each country to consume beyond its production possibilities frontier (PPF).

Problem 3: Terms of Trade and Trade Gains

Two countries, **Brazil** and **Canada**, produce **Coffee** and **Timber**. The labor productivity is as follows:

Country	Coffee (tons per worker)	Timber (tons per worker)
Brazil	12	4
Canada	8	6

(a) Find the opportunity cost of producing **1 ton of coffee** in each country.

(b) Suppose Brazil specializes in **coffee** and Canada specializes in **timber**. If they trade at a **terms of trade** of 1 ton of coffee = **0.5 tons of timber**, will both countries benefit? Explain why or why not.

Problem 4: Effect of Technological Change on Trade Patterns

Initially, two countries, **India** and **Germany**, produce **Steel** and **Textiles**. The productivity levels are:

Country	Steel (tons per worker)	Textiles (meters per worker)
India	5	20
Germany	10	30

(a) Determine each country's comparative advantage before any technological change.

(b) Suppose a technological improvement in Germany **doubles its steel productivity** but does not affect textiles. How does this affect Germany's comparative advantage?

(c) How would this change affect the **global pattern of trade** between India and Germany?